

Food-intake detection from AIM-2 sensor signals: literature review and model description

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1 Literature review

Objective measurement of eating behaviour matters for obesity, eating disorders and diabetes research, and self-report is known to be inaccurate and cannot capture in-meal behaviour such as eating rate or pauses (1). Wearable sensors that detect chewing, bites or hand gestures have therefore become a central approach, ranging from classical machine learning on hand-crafted features to end-to-end deep learning, with the aims of improving accuracy, reducing false detections and generalising to people outside the training set (2). Fontana et al. (3) introduced the Automatic Ingestion Monitor (AIM), combining jaw-motion, hand-gesture and accelerometer sensors, and reported 89.8% food-intake detection accuracy over 24 hours of free living. Its successor, AIM-2 (4), moved the sensors onto eyeglasses: a temporalis-muscle sensor and a 3-axis accelerometer at 128 Hz. It extracted 38 hand-crafted features per signal, selected nine by minimum-redundancy maximum-relevance, and classified 10 s epochs with a support vector machine, reaching F1 0.818 under leave-one-subject-out validation on 30 participants, as summarised in Table 1. These feature-engineered models work, but their performance is bounded by the features chosen by hand, and they are sensitive to noise and to variation between people.

Table 1: State-of-the-art methods for food-intake and chewing detection, and the two ECG studies whose methods transfer to this problem.

References	Techniques	Goals	Findings
Doulah et al. (4)	AIM-2; hand-crafted features, mRMR selection, SVM	Food-intake detection with passive food-image capture	F1 0.818 (LOSO, 30 participants); camera triggered only during eating
Papapanagiotou et al. (5)	1D CNN on raw in-ear audio	Chewing detection without hand-crafted features	Accuracy 0.980, F1 0.883
Kyritsis et al. (6)	CNN + LSTM, end to end on raw wrist IMU	Bite and meal detection in the wild	Bite F1 0.923 (LOSO)

References	Techniques	Goals	Findings
Ghosh and Sazonov (7)	Five deep networks (MLP, time-CNN, FCN, ResNet, Inception) on AIM-2 optical and accelerometer time series	Compare deep architectures for food-intake detection	ResNet best, balanced accuracy 93.47%
Ghosh et al. (8)	Wavelet scalogram of accelerometer, fused with food-image recognition	Reduce false positives in free living	Fused F1 80.77%; fusion cuts false positives
Stankoski et al. (9)	Selection of hard negative examples for training	Imbalanced data with imperfect labels	Person-independent F1 0.82
Wang et al. (10)	Temporal CNN + multi-head self-attention	Eating speed from wrist IMU, towards free living	61 participants, 513 h; eating-speed MAPE 0.110
Vedovelli et al. (11)	Transformer encoder and other models on smartwatch sensors	Eating-episode recognition	Transformer: sensitivity 0.691, specificity 0.504 (LOSO)
Hannun et al. (12)	34-layer 1D CNN on raw single-lead ECG	End-to-end arrhythmia classification	F1 0.837, above the average cardiologist (0.780)
Ikram et al. (13)	Transformer on MIT-BIH beats	Five-class arrhythmia classification	Accuracy 97.1%; split not at patient level

1.1 From hand-crafted features to learned representations

To address the limits of hand-crafted features, several groups trained networks directly on the raw signal. Papapanagiotou et al. (5) detected chewing with a 1D convolutional network applied to raw in-ear audio windows, and Kyritsis et al. (6) trained a CNN feature extractor and an LSTM jointly on raw wrist inertial data, reaching a bite-detection F1 of 0.923 under leave-one-subject-out evaluation. On AIM-2 itself, Ghosh and Sazonov (7) compared five deep architectures on the optical and accelerometer time series and found the residual network best, with a balanced accuracy of 93.47%. The same pattern holds outside eating: the fully convolutional network of Wang et al. (14), three blocks of convolution, batch normalisation and ReLU, is a strong baseline across time-series benchmarks.

The closest analogue to this problem is electrocardiography. Hannun et al. (12) classified arrhythmia with a 34-layer 1D CNN applied end to end to raw single-lead ECG at 200 Hz, and exceeded the average cardiologist (F1 0.837 against 0.780). ECG and the AIM-2 signals share a sampling rate of the same order, a quasi-periodic structure (heart beats at about 1 Hz, chews at 0.94–2.17 Hz (15)) and the need to recognise short events inside longer recordings. This motivates the first and most important design choice in this work: the model reads the time series, not a spectrum. Ghosh et al. (8) took the opposite route, converting accelerometer signals to a wavelet scalogram image. A magnitude spectrum says which rhythms are present in a window but discards when they occur, which is the information that separates a bite from background motion; a scalogram keeps a time axis, but at a resolution fixed by the transform rather than learned from the data.

1.2 Attention and Transformers

Attention mechanisms let a network weight the parts of its input that carry evidence. Squeeze-and-excitation (16) gates feature channels with a small learned function of their global average, at negligible cost. Attention-based pooling (17) treats a window as a bag of instances and learns which instances determine its label; for eating, the instances are time steps and a single bite may be the deciding one. Transformers extend this to attention between all time steps. Ikram et al. (13) applied a Transformer to five-class arrhythmia classification on MIT-BIH, reporting 97.1% accuracy. The reported 80/20 split is of preprocessed samples rather than of patients, so the figure measures performance on new beats from known patients, not on new patients. In eating detection, Wang et al. (10) combined a temporal CNN with multi-head self-attention to measure eating speed from wrist IMU data on 61 participants, while Vedovelli et al. (11) found that a Transformer encoder on smartwatch data under leave-one-subject-out gave the highest sensitivity (0.691) but a specificity of only 0.504, which is the false-positive problem in its clearest form.

Despite their promise, Transformers carry many more parameters than the attention layers above, and whether they help at a given data size is an empirical question rather than a given. This work therefore uses the two light attention mechanisms by default and tests a Transformer over the convolutional features as a separate experiment, on the same folds.

1.3 Supervision, augmentation and negative data

Small annotated datasets call for more supervision per example and for more examples per recording. Multi-task learning (18) improves generalisation by training a shared representation on related tasks; here, counting chews is the related task, since the ground truth already contains chew annotations. Augmentation of wearable sensor data raised accuracy from 77.54% to 86.88% in Parkinson’s disease monitoring (19); overlapping windows are the simplest form of it. Window length itself trades resolution for context, and has a measurable effect on recognition (20).

False positives are the persistent weakness of eating detectors: many non-eating activities produce jaw and head motion. Stankoski et al. (9) reduced them by deliberately selecting hard negative examples for training, and Ghosh et al. (8) by fusing the sensor classifier with food-image recognition. Finally, evaluation practice varies widely across the field (2). Saeb et al. (21) showed that record-wise cross-validation, where windows from one person appear in both training and test sets, can massively overestimate accuracy, and that subject-wise validation is the one that matches the use case. The leave-one-subject-out protocol used by Doulah et al. (4) and Diou et al. (1) is adopted here for the same reason.

1.4 Contribution of this work

- A compact convolutional model with channel attention, temporal attention pooling and an auxiliary chew-count head, reading raw AIM-2 optical and accelerometer signals directly from the ETS database, evaluated leave-one-subject-out.
- A controlled ablation of each design decision, one at a time on identical participant-level folds, separating the components that carry the performance from those that do not.
- Negative results reported with their mechanism: temporal median smoothing and context windows without a target marker both lower F1, because meals at 8 s resolution are punctuated rather than contiguous.

- A measured check of sensor-annotation alignment on every session, replacing an assumed correction with a verified one.

Despite the progress in deep learning for intake monitoring, several challenges persist: limited annotated data, variation between participants in sensor fit and chewing style, false positives from non-eating jaw and head motion, and inconsistent evaluation that makes published results hard to compare. Addressing these requires models that generalise to people they have not seen, evaluated in the way they would be used.

2 Model description

The model is a compact one-dimensional convolutional network that reads the raw AIM-2 sensor signals and outputs the probability that an 8 s window contains eating (Figure 1). It has four parts. A convolutional feature extractor learns its own filters from the time series instead of receiving hand-crafted features or a spectrum. A squeeze-and-excitation gate reweights the learned feature channels in each window. An attention pooling layer weights the time steps that carry the evidence. Finally, the output has two heads: one classifies eating, and one counts chews and is used only during training. The network has 47,282 parameters at inference (Table 2). Table 3 lists every design decision with its reason and the literature it rests on.

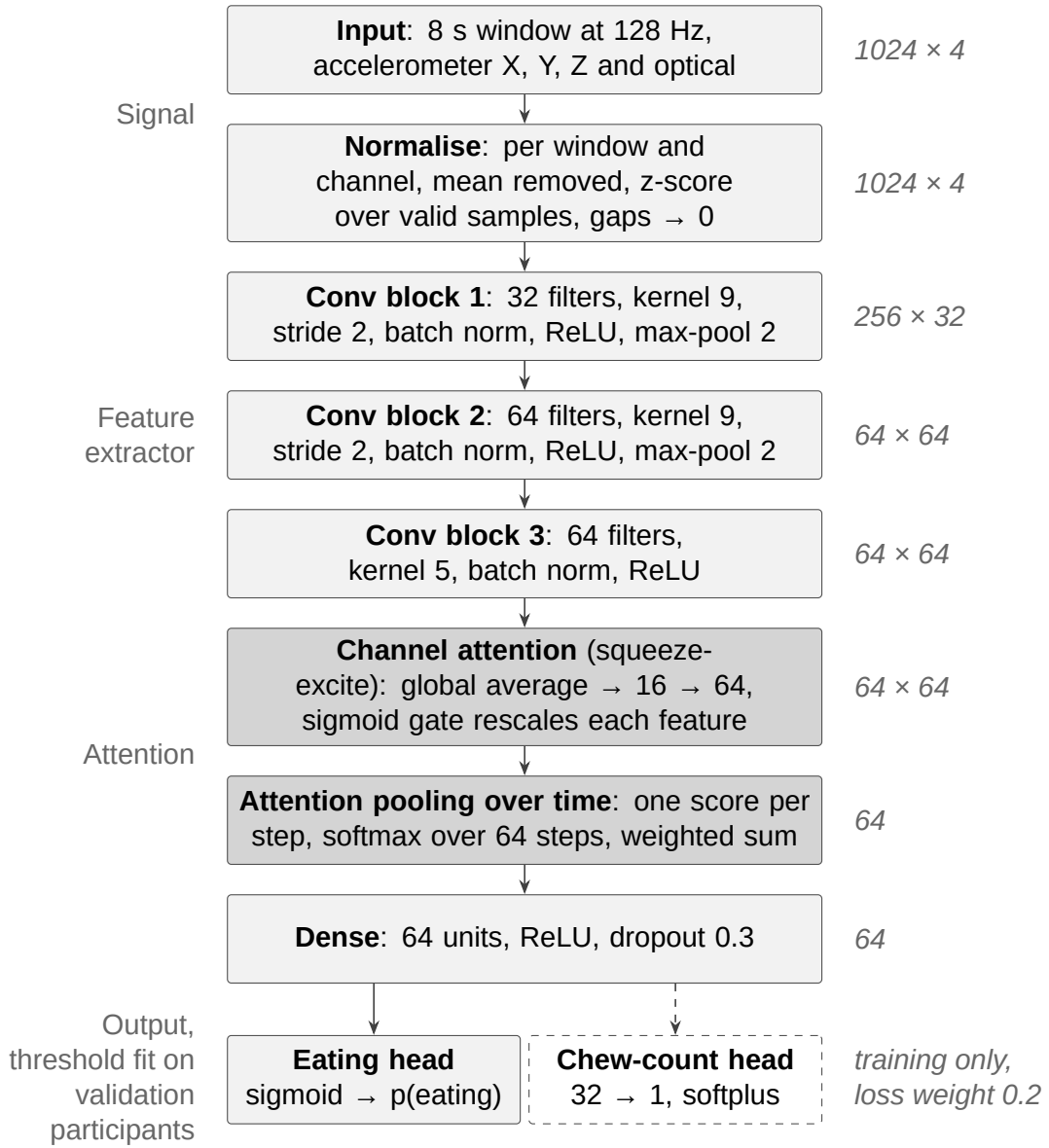


Figure 1: Model architecture. Shaded blocks are the two attention mechanisms. The dashed chew-count head is trained jointly and discarded at inference. Output shapes are time steps $\times$ features.

Table 2: Layer dimensions. An 8 s window at 128 Hz is 1024 samples; each of the 64 output time steps covers about 0.9 s of signal at 125 ms spacing.

Layer	Configuration	Output	Parameters
Input	Acc X, Y, Z, Optical	1024 × 4	–
Conv block 1	32 × k9, stride 2, BN, max-pool 2	256 × 32	1,312
Conv block 2	64 × k9, stride 2, BN, max-pool 2	64 × 64	18,752
Conv block 3	64 × k5, BN	64 × 64	20,800
Channel attention	64 → 16 → 64, sigmoid	64 × 64	2,128
Attention pooling	1 × 1 conv, softmax over time	64	65
Dense	64, ReLU, dropout 0.3	64	4,160
Eating head	1, sigmoid	1	65
<i>Total at inference</i>			47,282
Chew-count head	32 → 1, softplus (training only)	1	2,113

Table 3: Complete description of the food-intake detection model. Every design decision, the reason for it, and the literature it is grounded in.

#	Component	Design	Why	Literature
<i>Data</i>				
1	Sensor	AIM-2 eyeglass sensor: 3-axis accelerometer and optical temporalis-muscle sensor, 128 Hz, stored as 8 s packets in the ETS database	Chewing moves the temporalis muscle and the head; both channels carry it, with different artefacts. The original AIM-2 used a flex sensor where this one has the optical sensor	Doulah et al. (4), Ghosh and Sazonov (7)
2	Labels	Eating = bout OR bite annotation at 10 Hz. Window is eating if more than half its annotated samples are eating. Unrecorded time (−1) is dropped, never treated as non-eating	Treating unannotated time as negative would bury the real labels and invent easy negatives	(this work)
3	Alignment	No time shift. Sensor/annotation lag measured by cross-correlating the optical chewing envelope with the labels: median +0.10 s, all within 0.2 s on the 20 sessions measured first (to be repeated on all 61)	A correction must be measured, not assumed; a residual lag of one packet would corrupt every label edge	(this work)

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#	Component	Design	Why	Literature
4	Window	8 s (1024 samples), non-overlapping at evaluation	7 to 17 chew cycles at typical chewing rates (0.94–2.17 Hz); matches the packet size. Window length has a measurable effect and is tested at 2–16 s	Po et al. (15), Banos et al. (20)
<i>Input</i>				
5	Representation	Raw time series, 1024 samples $\times$ 4 channels, no hand-crafted features and no spectrum	A magnitude spectrum keeps which rhythms are present but discards when they occur; a bite is a transient and chewing an envelope. Largest single effect in the preliminary ablation	Ghosh and Sazonov (7), Hannun et al. (12), Wang et al. (14)
6	Normalisation	Per window, per channel: mean removed, z-scored over valid samples only; missing samples set to 0 after scaling	Optical baseline varies by participant and fit on the face; DC level is not information about chewing. A literal -1 against counts near 3000 injects a step at every gap	Ismail Fawaz et al. (22)
<i>Network</i>				
7	Convolutional trunk	Three 1D conv blocks: $32@k9/s2$, $64@k9/s2$, $64@k5$, each with batch norm and ReLU, max-pool 2 after the first two. Output 64 time steps $\times$ 64 features	Learns its own band-pass filters from data. Each output step covers about 0.9 s of signal (one chew cycle) at 125 ms spacing	Ghosh and Sazonov (7), Wang et al. (14)
8	Channel attention	Squeeze-and-excitation: global average $\rightarrow$ dense 16 (ReLU) $\rightarrow$ dense 64 (sigmoid) $\rightarrow$ rescale each feature channel	Optical carries chewing; accelerometers carry motion that is sometimes signal and often artefact. Reweights per window at negligible cost	Hu et al. (16)
9	Temporal pooling	Attention pooling: one score per time step (1×1 conv), softmax over the 64 steps, weighted sum	Global averaging dilutes a 1–2 s bite across 8 s. Attention weights the informative steps, as in multiple-instance learning	Ilse et al. (17)
10	Classifier	Dense 64 (ReLU), dropout 0.3, sigmoid output	Small head; about 47 k parameters in total at inference	–

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#	Component	Design	Why	Literature
11	Auxiliary head	Chew-count regression (dense 32 $\rightarrow$ 1, softplus), target scaled by its 99th percentile. Used in training only	One label bit per window is thin supervision; counting chews asks the shared trunk to represent the chewing rhythm	Caruana (18)
<i>Training</i>				
12	Loss	Binary cross-entropy (weight 1.0) + chew MSE (weight 0.2); class-balanced sample weights computed per fold	Auxiliary loss is a regulariser, not the objective. Eating fraction ranges 0–86% by session, so the rare class changes with the fold	Caruana (18)
13	Augmentation	50% overlapping windows (4 s hop) at training time only	Roughly doubles the training windows at no collection cost; evaluation stays on the non-overlapping grid so no second is scored twice	Um et al. (19)
14	Optimisation	Adam, learning rate 0.001, batch 64, up to 80 epochs, early stopping (patience 12) and learning-rate halving on validation loss	Standard; validation participants are held out from training, never random windows	–
<i>Evaluation</i>				
15	Validation	Leave-one-subject-out: 43 folds, each holding out one participant with all of their sessions. Inner validation on further held-out participants	Windows of one meal are near-duplicates; any split below participant level leaks and overstates performance	Diou et al. (1), Saeb et al. (21)
16	Threshold	Chosen on the inner validation participants at the centre of the F1 plateau; never on the test participant	The F1 curve is flat near its peak, so the exact argmax on two participants is noise	–
17	Post-processing	None. Median smoothing over 3 windows was tested and rejected (F1 -0.079)	Meals are punctuated by pauses; at 8 s the label genuinely switches, and smoothing erases short real bouts	(this work)
18	Metrics	F1 and balanced accuracy lead; accuracy, precision, recall reported alongside, pooled over all held-out windows	Class balance differs by participant; accuracy alone rewards predicting the majority class	Bell et al. (2)

2.1 Dataset and current performance

The dataset is 61 annotated eating sessions (breakfast, lunch, dinner and snack) from 43 participants, 2 to 55 minutes each: 24.2 hours of annotated recording, of which 9.6 hours is eating. The eating fraction ranges from 0% to 86% by session. Validation is leave-one-subject-out: 43 folds, each holding out one participant with all of their sessions.

In a preliminary participant-level 5-fold cross-validation, pooled over all 10,891 held-out non-overlapping windows, the model reaches F1 0.858, accuracy 0.891, balanced accuracy 0.881, precision 0.884 and recall 0.834, with 32 false alarms per hour of non-eating. Per-participant F1 has mean 0.835 (standard deviation 0.136, median 0.878); the spread between people is the honest measure of how the model generalises to someone new, and it is larger than any difference between design choices seen so far. Leave-one-subject-out results will replace these figures.

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